Analysis Report

Maternal Mortality Ratio Statistical Analysis

World Bank Dataset (API_SH.STA.MMRT_DS2)

This notebook presents comprehensive statistical analysis of global Maternal Mortality Ratio (MMR) data from 1960-2025, including: - Descriptive statistics and data exploration - Inferential statistical tests - Temporal trend analysis - Regional comparisons - Visualizations and charts

```
Dataset loaded successfully!
Shape: (266, 66)
Countries/Regions: 266
Years: 66 (1960 - 2025)

First 10 rows and columns:
                                        1960  1961  1962  1963  1964  1965  \
                          Country_Code
Aruba                     ABW            NaN   NaN   NaN   NaN   NaN   NaN
Africa Eastern and Southern AFE          NaN   NaN   NaN   NaN   NaN   NaN
Afghanistan               AFG            NaN   NaN   NaN   NaN   NaN   NaN
Africa Western and Central  AFW          NaN   NaN   NaN   NaN   NaN   NaN
Angola                    AGO            NaN   NaN   NaN   NaN   NaN   NaN
Albania                   ALB            NaN   NaN   NaN   NaN   NaN   NaN
Andorra                   AND            NaN   NaN   NaN   NaN   NaN   NaN
Arab World                ARB            NaN   NaN   NaN   NaN   NaN   NaN
United Arab Emirates      ARE            NaN   NaN   NaN   NaN   NaN   NaN
Argentina                 ARG            NaN   NaN   NaN   NaN   NaN   NaN

                                        1966  1967  1968  1969
                          Country_Code
Aruba                     ABW            NaN   NaN   NaN   NaN
Africa Eastern and Southern AFE          NaN   NaN   NaN   NaN
Afghanistan               AFG            NaN   NaN   NaN   NaN
Africa Western and Central  AFW          NaN   NaN   NaN   NaN
Angola                    AGO            NaN   NaN   NaN   NaN
Albania                   ALB            NaN   NaN   NaN   NaN
Andorra                   AND            NaN   NaN   NaN   NaN
Arab World                ARB            NaN   NaN   NaN   NaN
United Arab Emirates      ARE            NaN   NaN   NaN   NaN
Argentina                 ARG            NaN   NaN   NaN   NaN
```

1. Data Exploration and Missing Data Analysis

```
======================================================================
MISSING DATA ANALYSIS
======================================================================

Missing values per year (first and last 10 years):
1960    266
1961    266
1962    266
1963    266
1964    266
1965    266
1966    266
1967    266
1968    266
1969    266
dtype: int64
...
2015     24
2016     24
2017     24
2018     24
2019     24
2020     24
2021     24
2022     24
2023     24
2024    266
dtype: int64


Total missing values: 7852
Overall missing percentage: 45.41%


Countries with most data completeness (top 10):
Countries with complete data (no missing values): 0

Top 10 countries by data completeness:
                           Country  Completeness_%
(Africa Eastern and Southern, AFE)            60.0
                (Afghanistan, AFG)            60.0
 (Africa Western and Central, AFW)            60.0
                     (Angola, AGO)            60.0
                    (Albania, ALB)            60.0
                    (Andorra, AND)            60.0
                 (Arab World, ARB)            60.0
       (United Arab Emirates, ARE)            60.0
```

```
                  (Argentina, ARG)            60.0
                    (Armenia, ARM)            60.0
```

2. Descriptive Statistics - Overall Summary

```
======================================================================
DESCRIPTIVE STATISTICS - ALL MATERNAL MORTALITY RATIO VALUES
======================================================================

Count: 9438
Mean: 245.80
Median: 86.00
Mode: 7.00
Standard Deviation: 384.23
Variance: 147635.84
Minimum: 1.00
Maximum: 8045.00
Range: 8044.00
Q1 (25th percentile): 22.00
Q3 (75th percentile): 342.00
IQR: 320.00
Skewness: 5.8863
Kurtosis: 80.5697

======================================================================
SUMMARY STATISTICS BY YEAR (Latest 10 years with data)
======================================================================
      count        mean         std  min    25%   50%     75%     max
2015  242.0  167.772727  231.812259  1.0  15.00  70.0  222.00  1436.0
2016  242.0  161.165289  227.997501  1.0  15.00  67.0  214.00  1629.0
2017  242.0  155.917355  224.855309  1.0  15.25  63.0  200.00  1662.0
2018  242.0  147.190083  198.533872  1.0  14.25  66.0  198.75  1118.0
2019  242.0  144.896694  207.654801  1.0  14.25  65.0  190.75  1650.0
2020  242.0  145.293388  192.167784  2.0  20.00  74.0  194.00  1296.0
2021  242.0  162.533058  189.813976  2.0  25.25  98.0  218.25  1127.0
2022  242.0  134.549587  171.272197  1.0  17.00  71.5  181.75  1016.0
2023  242.0  126.264463  165.080429  1.0  14.00  61.0  173.00   993.0
2024    0.0         NaN         NaN  NaN    NaN   NaN     NaN     NaN
```

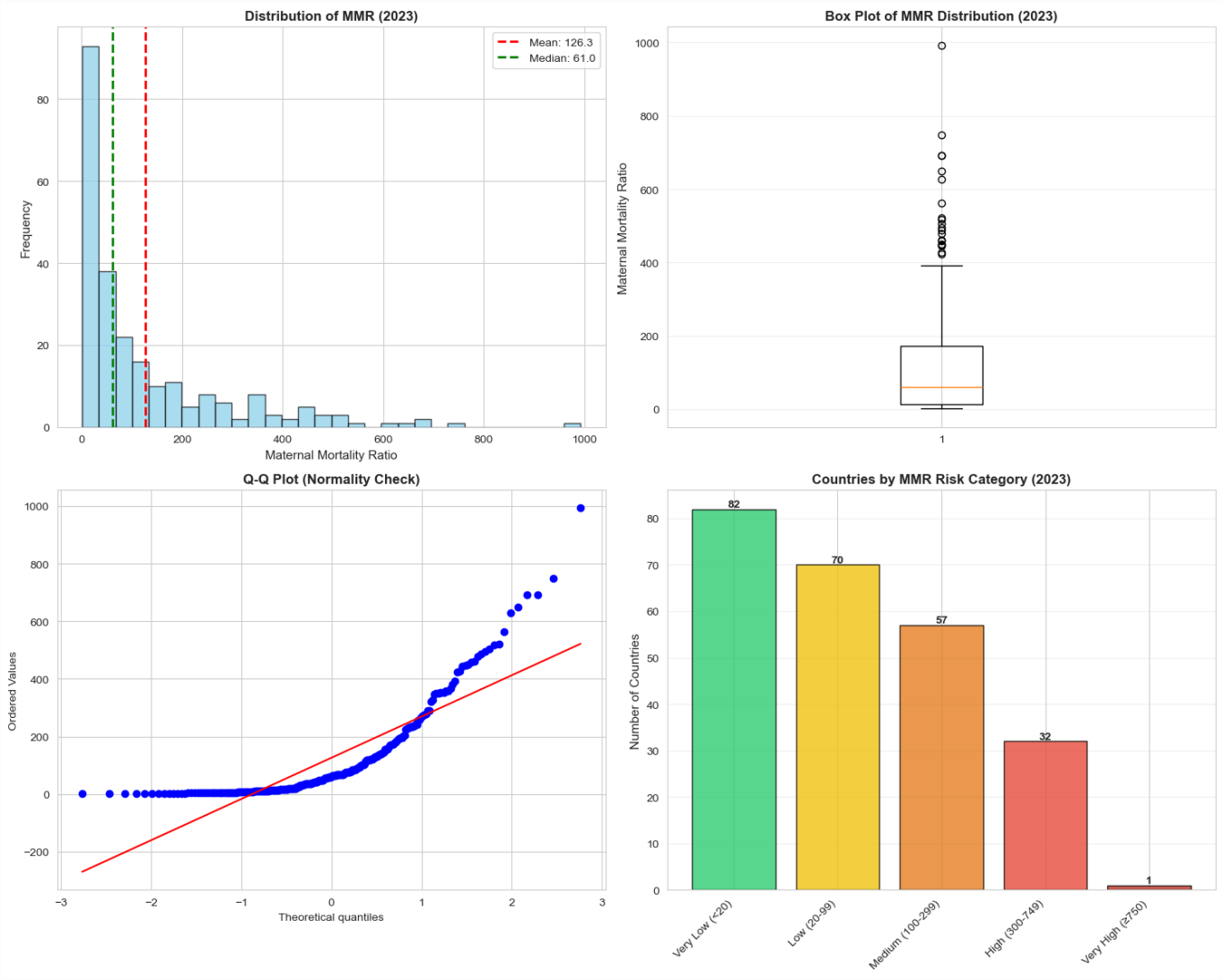
```
================================================================================
FREQUENCY DISTRIBUTION - MMR RISK CATEGORIES (2023)
================================================================================

Risk Level           Count      Percentage
----------------------------------------
Very Low (<20)       82         33.9%
Low (20-99)          70         28.9%
Medium (100-299)     57         23.6%
High (300-749)       32         13.2%
Very High (≥750)     1          0.4%

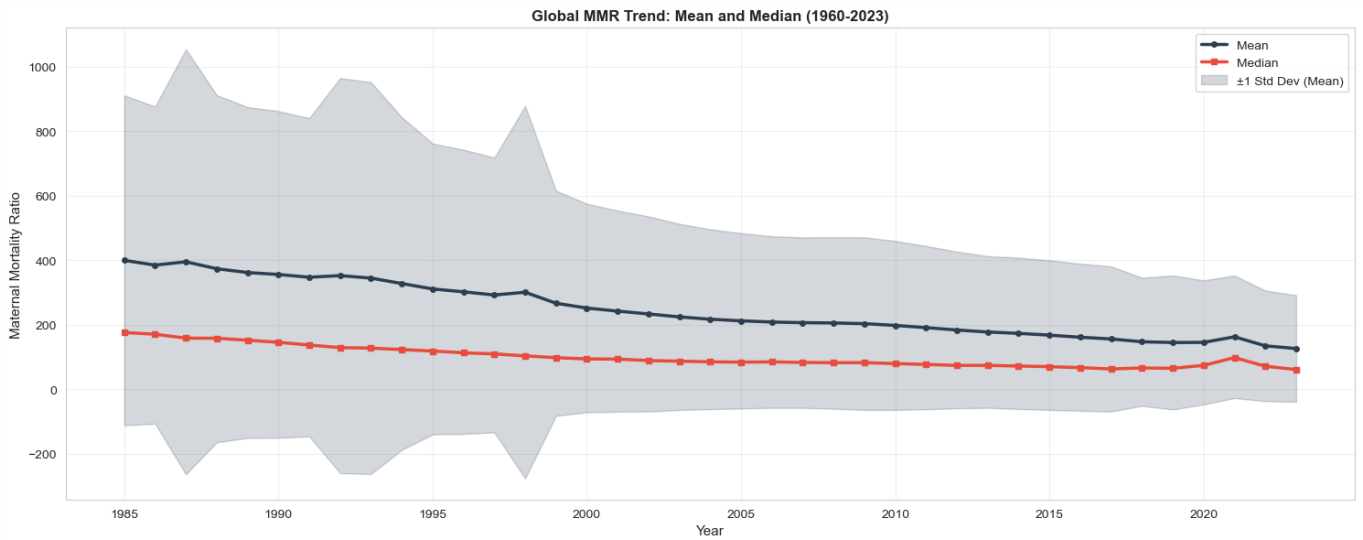
================================================================================
TOP 10 COUNTRIES WITH HIGHEST MMR (2023)
================================================================================
 1. Nigeria                                  993
 2. Chad                                     748
 3. Central African Republic                 692
 4. South Sudan                              692
 5. Africa Western and Central               650
 6. Liberia                                  628
 7. Somalia, Fed. Rep.                       563
 8. Afghanistan                              521
 9. Benin                                    518
10. Guinea-Bissau                            505

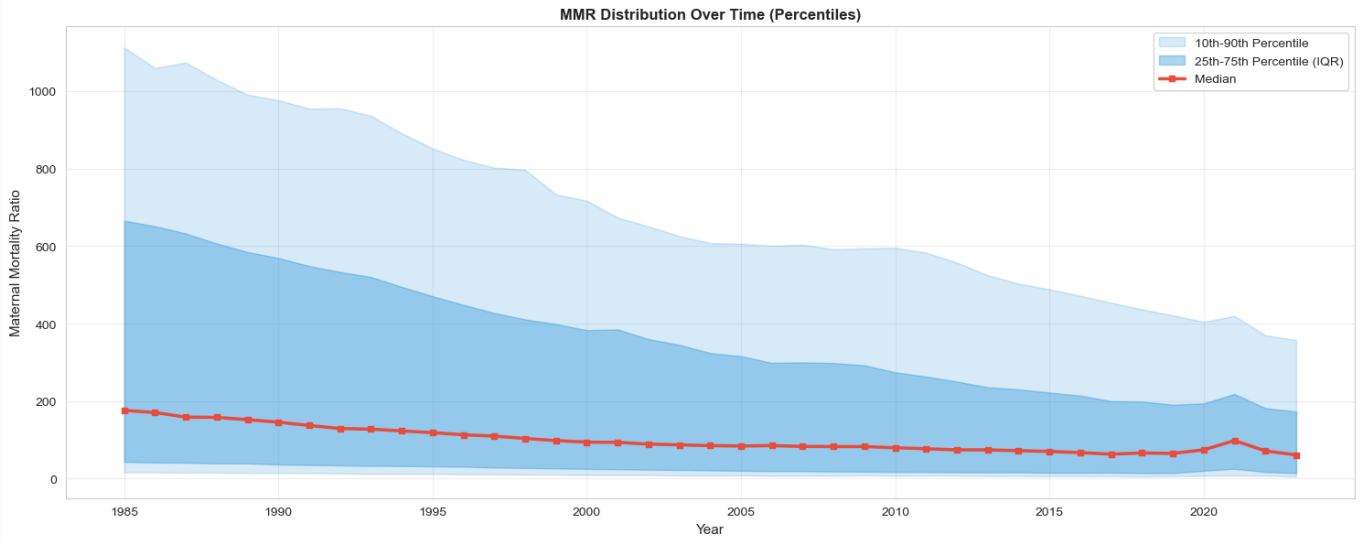
================================================================================
TOP 10 COUNTRIES WITH LOWEST MMR (2023)
================================================================================
 1. Belarus                                    1
 2. Norway                                     1
 3. Australia                                  2
 4. Israel                                     2
 5. Poland                                     2
 6. United Arab Emirates                       3
 7. Czechia                                    3
 8. Spain                                      3
 9. Croatia                                    3
10. Iceland                                    3
```

3. Data Visualization - Distributions and Trends




```
Visualizations created successfully!
```






```
Trend analysis complete!
```

4. Inferential Statistics - Correlation and Trend Analysis

```
======================================================================
CORRELATION ANALYSIS - TIME SERIES
======================================================================


Correlation Matrix - Every 10 Years:
      1960  1970  1980      1990      2000      2010      2020
1960   NaN   NaN   NaN       NaN       NaN       NaN       NaN
1970   NaN   NaN   NaN       NaN       NaN       NaN       NaN
1980   NaN   NaN   NaN       NaN       NaN       NaN       NaN
1990   NaN   NaN   NaN  1.000000  0.895987  0.772417  0.753305
2000   NaN   NaN   NaN  0.895987  1.000000  0.917309  0.893060
2010   NaN   NaN   NaN  0.772417  0.917309  1.000000  0.931991
2020   NaN   NaN   NaN  0.753305  0.893060  0.931991  1.000000

======================================================================
INDIVIDUAL COUNTRY TRENDS (Year vs MMR)
======================================================================


Countries with SIGNIFICANT DECLINING MMR (Improvement):
Top 10 countries with steepest improvement:

                                       Country  Correlation      P-Value
                                  (Guinea, GIN)    -0.999008 1.300789e-51
                                (Ethiopia, ETH)    -0.995702 7.637755e-40
                                 (Lao PDR, LAO)    -0.993307 2.714233e-36
  (Heavily indebted poor countries (HIPC), HPC)    -0.992084 5.999487e-35
              (Early-demographic dividend, EAR)    -0.990011 4.359150e-33
                                    (Niger, NER)    -0.988903 3.020381e-32
                          (IDA & IBRD total, IBT)    -0.988655 4.539973e-32
(Fragile and conflict affected situations, FCS)    -0.987776 1.790962e-31
                               (South Asia, SAS)    -0.986796 7.401449e-31
                                (IDA total, IDA)    -0.986791 7.451322e-31


Countries with SIGNIFICANT INCREASING MMR (Worsening):
              Country  Correlation      P-Value
        (Canada, CAN)     0.926036 3.087349e-17
       (Jamaica, JAM)     0.903300 3.613063e-15
 (Venezuela, RB, VEN)     0.841703 1.914696e-11
 (North America, NAC)     0.805858 6.048686e-10
 (United States, USA)     0.803467 7.422651e-10
(United Kingdom, GBR)     0.609229 3.853821e-05
         (Nauru, NRU)     0.605769 4.381327e-05
  (Bahamas, The, BHS)     0.517250 7.465408e-04
        (Cyprus, CYP)     0.325398 4.324078e-02
```

```
================================================================================
SUMMARY STATISTICS OF TRENDS
================================================================================
Countries with significant declining trend: 216 (89.3%)
Countries with significant increasing trend: 9 (3.7%)
Mean trend correlation: -0.7477
Median trend correlation: -0.9163
```

```
Analysis window set to 1985-2023
Late 20th century: 1985-1999 (3630 observations)
21st century: 2000-2023 (5808 observations)
```

```
======================================================================
CHI-SQUARE TEST - ASSOCIATION ANALYSIS
======================================================================


Contingency Table - Late 20th Century vs 21st Century:
Risk_Category                  High (300-749)  Low (20-99)  Medium (100-299)  Very
High (≥750)  Very Low (<20)
Period

21st Century (2000-2023)                1076         1736
1224               214            1558
Late 20th Century (1985-1999)            711         1032
639               596             652


Chi-Square Test Results:
Chi-Square Statistic: 513.6086
P-Value: 7.6312e-110
Degrees of Freedom: 4
Interpretation: Significant association between period and MMR risk category (α =
0.05)


Expected Frequencies:
Risk_Category                  High (300-749)  Low (20-99)  Medium (100-299)  Very
High (≥750)  Very Low (<20)
Period

21st Century (2000-2023)             1099.69      1703.38
1146.46            498.46          1360.0
Late 20th Century (1985-1999)          687.31      1064.62
716.54            311.54           850.0
```

```
================================================================================
T-TESTS - COMPARING MMR BETWEEN PERIODS
================================================================================

1. T-Test: Late 20th Century (1985-1999) vs 21st Century (2000-2023)
   Mean (Late 20th Century (1985-1999)): 340.91
   Mean (21st Century (2000-2023)): 186.36
   T-Statistic: 16.7725
   P-Value: 2.2889e-61
   Result: Significant difference


Effect Size (Cohen's d):
Late 20th Century (1985-1999) vs 21st Century (2000-2023): 0.4102
```

```
======================================================================
LINEAR REGRESSION ANALYSIS - GLOBAL MMR TREND
======================================================================


Global MMR Trend Regression (All Countries Combined):
Slope: -7.2623 MMR per year
Intercept: 14799.49
R-squared: 0.9613
Correlation Coefficient (r): -0.9804
P-Value: 1.0067e-27
Standard Error: 0.2397

Interpretation: MMR decreases by 7.2623 per year
Statistical Significance: Yes (p < 0.05)

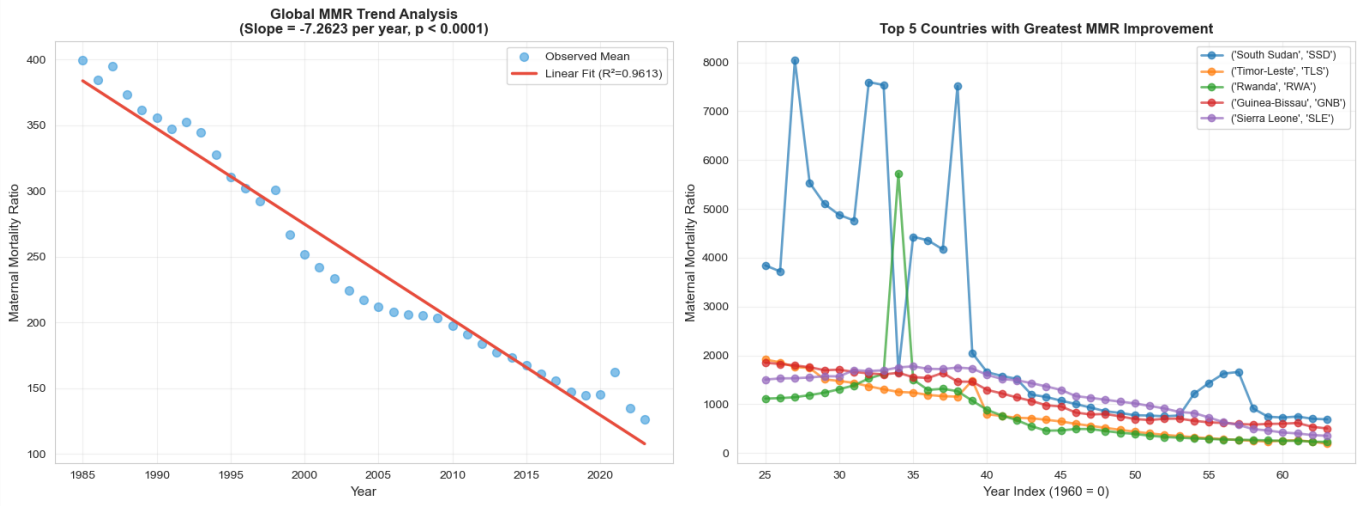

Projected MMR Values (based on linear trend):
  2024: 100.6
  2025: 93.3
  2026: 86.0
  2027: 78.8
  2028: 71.5


======================================================================
REGRESSION ANALYSIS - TOP 10 IMPROVING COUNTRIES
======================================================================

Top 10 Countries with Most Rapid MMR Decline:
             Country       Slope  R-Squared      P-Value Significance
  (South Sudan, SSD) -153.431984   0.579371 1.852686e-08          Yes
  (Timor-Leste, TLS)  -46.018623   0.926652 1.381349e-22          Yes
       (Rwanda, RWA)  -45.984008   0.323346 1.588417e-04          Yes
(Guinea-Bissau, GNB)  -40.359312   0.943331 1.158226e-24          Yes
 (Sierra Leone, SLE)  -38.640688   0.842593 1.972456e-16          Yes
  (Afghanistan, AFG)  -35.105870   0.942914 1.326721e-24          Yes
   (Mozambique, MOZ)  -34.973077   0.895830 9.241357e-20          Yes
      (Burundi, BDI)  -34.293522   0.873169 3.567885e-18          Yes
      (Liberia, LBR)  -34.230567   0.698261 3.642467e-11          Yes
      (Eritrea, ERI)  -34.203644   0.871979 4.243467e-18          Yes


Top 10 Countries with Most Rapid MMR Increase:
                        Country    Slope  R-Squared      P-Value Significance
(Central African Republic, CAF) 4.251417   0.038144 2.334519e-01           No
          (Venezuela, RB, VEN) 3.697976   0.708463 1.914696e-11          Yes
```

```
                                (Lesotho, LSO) 2.888057   0.015002 4.575945e-01          No
                                (Jamaica, JAM) 2.247166   0.815951 3.613063e-15         Yes
                                  (Nauru, NRU) 2.244130   0.366956 4.381327e-05         Yes
                            (Bahamas, The, BHS) 0.887247   0.267548 7.465408e-04         Yes
                                (Zimbabwe, ZWE) 0.778745   0.013557 4.802585e-01          No
                                  (Cyprus, CYP) 0.540486   0.105884 4.324078e-02         Yes
                           (United States, USA) 0.278947   0.645560 7.422651e-10         Yes
                          (North America, NAC) 0.272267   0.649408 6.048686e-10         Yes
```




```
Regression visualization complete!
```

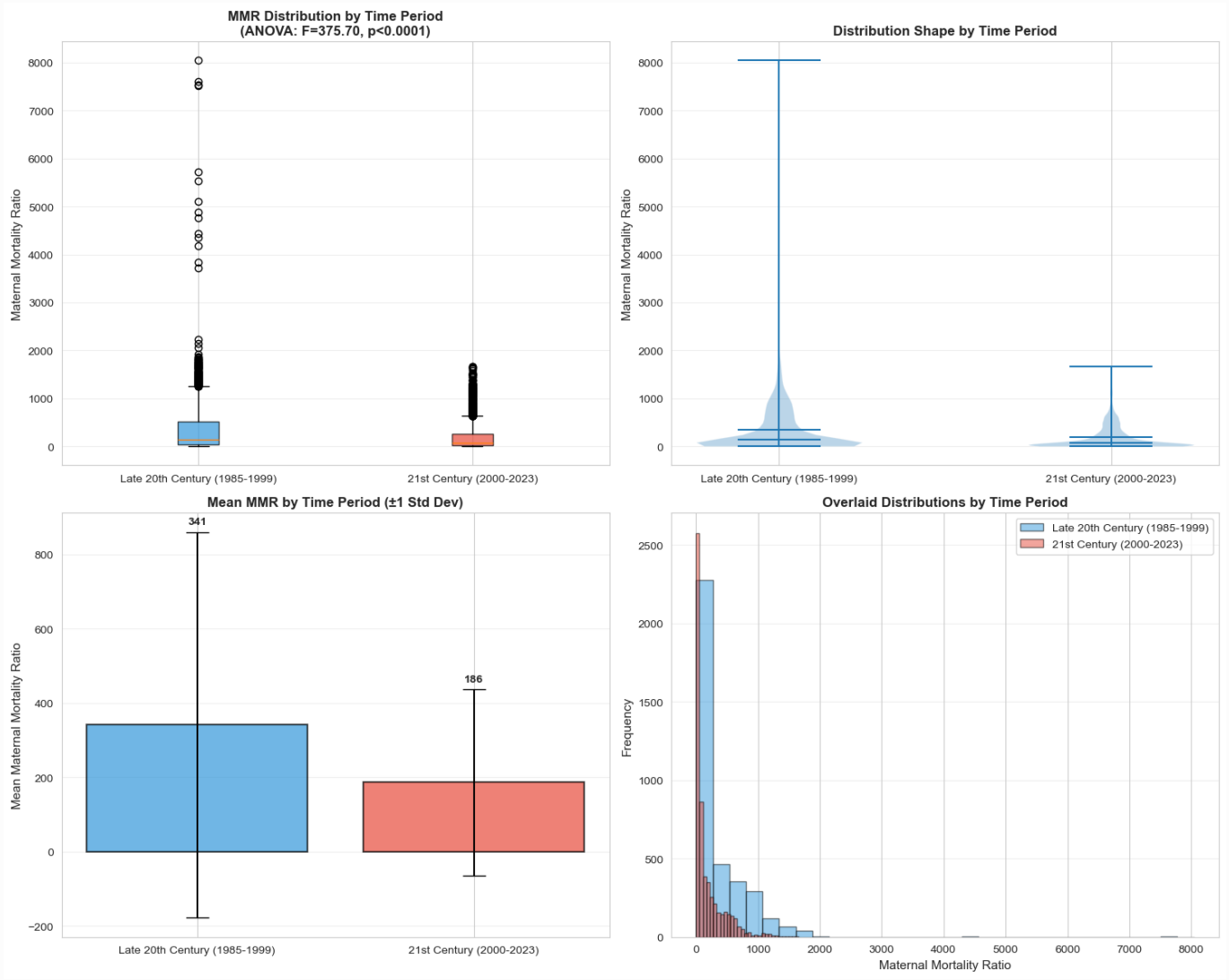
```
================================================================================
ANOVA TEST - COMPARING MMR ACROSS PERIODS
================================================================================

One-Way ANOVA Results:
F-Statistic: 375.7039
P-Value: 4.1788e-82
Interpretation: Significant difference in MMR across the two periods
Effect Size (Eta-squared): 0.0404
Interpretation: Small effect


Summary Statistics by Time Period:

Late 20th Century (1985-1999):
  Mean: 340.91
  Median: 130.00
  Std Dev: 518.64
  Min: 2.00, Max: 8045.00
  N: 3630

21st Century (2000-2023):
  Mean: 186.36
  Median: 75.00
  Std Dev: 250.21
  Min: 1.00, Max: 1662.00
  N: 5808
```




```
ANOVA comparison visualization complete!
```

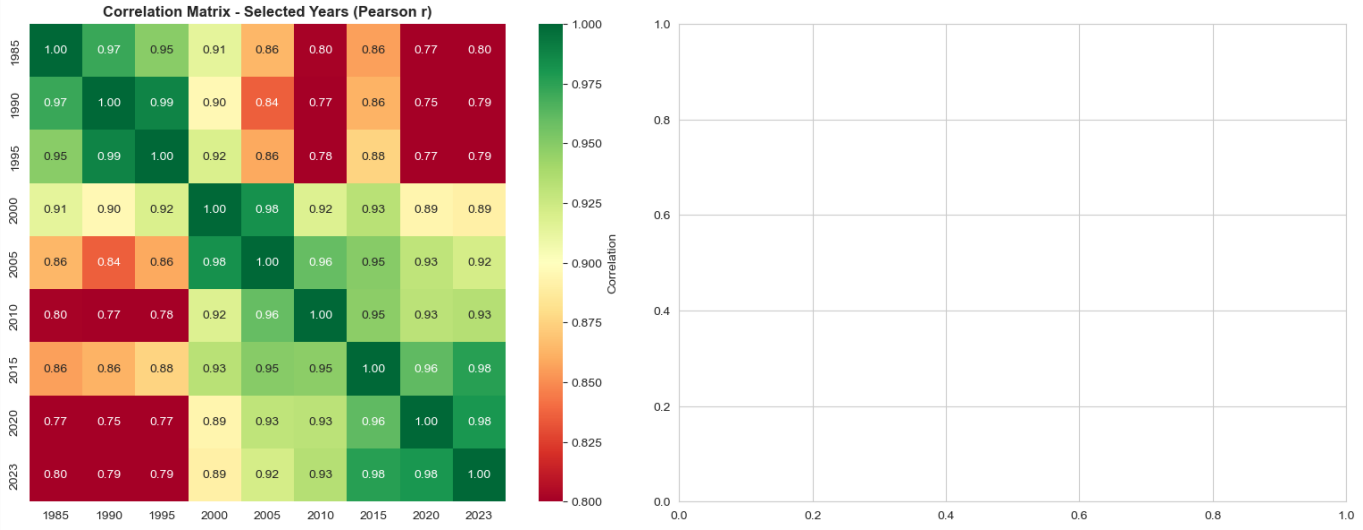
```
================================================================================
SPEARMAN RANK CORRELATION ANALYSIS
================================================================================


Spearman Rank Correlation (Global MMR vs Year):
Correlation Coefficient (ρ): -0.9962
P-Value: 9.8418e-41
Interpretation: Significant rank correlation between year and MMR


Comparison of Correlation Methods:
Pearson r: -0.9804 (linear relationship)
Spearman ρ: -0.9962 (monotonic relationship)
Difference: 0.0157


================================================================================
INTER-YEAR CORRELATION MATRIX (Selected Years)
================================================================================

Pearson Correlation Matrix:
       1985   1990   1995   2000   2005   2010   2015   2020   2023
1985  1.000  0.971  0.949  0.914  0.864  0.801  0.856  0.767  0.797
1990  0.971  1.000  0.988  0.896  0.836  0.772  0.862  0.753  0.785
1995  0.949  0.988  1.000  0.919  0.859  0.784  0.875  0.769  0.793
2000  0.914  0.896  0.919  1.000  0.982  0.917  0.932  0.893  0.890
2005  0.864  0.836  0.859  0.982  1.000  0.960  0.949  0.930  0.921
2010  0.801  0.772  0.784  0.917  0.960  1.000  0.948  0.932  0.934
2015  0.856  0.862  0.875  0.932  0.949  0.948  1.000  0.961  0.976
2020  0.767  0.753  0.769  0.893  0.930  0.932  0.961  1.000  0.980
2023  0.797  0.785  0.793  0.890  0.921  0.934  0.976  0.980  1.000
```



```
Correlation heatmaps created successfully!
```

5. Summary Tables and Key Findings

```
======================================================================
SUMMARY TABLE 1: GLOBAL STATISTICS ACROSS TIME PERIODS
======================================================================

                  Time Period       Mean  Median    Std Dev  Min    Max  Count
Late 20th Century (1985-1999) 340.906612   130.0 518.641932  2.0 8045.0   3630
     21st Century (2000-2023) 186.359848    75.0 250.206731  1.0 1662.0   5808


======================================================================
SUMMARY TABLE 2: STATISTICAL TEST RESULTS
======================================================================

                          Test Statistic     P-Value Significant
     Linear Regression (Global)   -7.2623  1.0067e-27         Yes
   Pearson Correlation (Global)   -0.9804  1.0067e-27         Yes
  Spearman Correlation (Global)   -0.9962  9.8418e-41         Yes
              ANOVA (2 Periods)  375.7039  4.1788e-82         Yes
     T-Test (Late 20th vs 21st)   16.7725  2.2889e-61         Yes
Chi-Square (Period vs Category)  513.6086 7.6312e-110         Yes


======================================================================
SUMMARY TABLE 3: TOP PERFORMERS - GREATEST MMR REDUCTION
======================================================================

             Country     Slope R-Squared  Data_Points
  (South Sudan, SSD) -153.4320    0.5794           39
  (Timor-Leste, TLS)  -46.0186    0.9267           39
       (Rwanda, RWA)  -45.9840    0.3233           39
(Guinea-Bissau, GNB)  -40.3593    0.9433           39
 (Sierra Leone, SLE)  -38.6407    0.8426           39
  (Afghanistan, AFG)  -35.1059    0.9429           39
   (Mozambique, MOZ)  -34.9731    0.8958           39
      (Burundi, BDI)  -34.2935    0.8732           39
      (Liberia, LBR)  -34.2306    0.6983           39
      (Eritrea, ERI)  -34.2036    0.8720           39
     (Ethiopia, ETH)  -32.1848    0.9914           39
  (Gambia, The, GMB)  -26.7739    0.9581           39
       (Malawi, MWI)  -25.6785    0.8119           39
        (Nepal, NPL)  -25.4638    0.8879           39
     (Lao PDR, LAO)  -24.6227    0.9867           39
```

```
================================================================================
NORMALITY TESTS - SHAPIRO-WILK TEST
================================================================================

Overall Data (Sample up to 5000 values):
  Test Statistic: 0.7335
  P-Value: 3.9316e-67
  Result: Non-normal distribution

By Time Period:

  Late 20th Century (1985-1999):
    Test Statistic: 0.6005
    P-Value: 4.0195e-68
    Result: Non-normal distribution

  21st Century (2000-2023):
    Test Statistic: 0.7307
    P-Value: 2.4402e-67
    Result: Non-normal distribution

================================================================================
DISTRIBUTION SHAPE ANALYSIS
================================================================================

Overall Data:
  Skewness: 5.8863 (Right-skewed)
  Kurtosis: 80.5697 (Leptokurtic (heavy-tailed))

By Time Period:

  Late 20th Century (1985-1999):
    Skewness: 5.4214
    Kurtosis: 57.7163

  21st Century (2000-2023):
    Skewness: 2.0920
    Kurtosis: 5.0860
```

6. Conclusions and Key Findings

```
================================================================================
COMPREHENSIVE STATISTICAL ANALYSIS SUMMARY
================================================================================


1. DESCRIPTIVE FINDINGS:
--------------------------------------------------------------------------------

   • Dataset: Maternal Mortality Ratio (MMR) per 100,000 live births
   • Valid analysis span: 1985-2023
   • Periods compared: Late 20th Century (1985-1999) vs 21st Century (2000-2023)
   • Overall Mean MMR (All periods): 245.8
   • Overall Median MMR: 86.0
   • Range: 1 - 8045

   Distribution Patterns:
   • The data shows positive skewness (5.89), indicating
     more countries with low MMR rates and fewer with very high rates
   • Heavy-tailed distribution (kurtosis: 80.57),
     suggesting presence of outliers

   Risk Category Distribution (2023):
   • Very Low (<20): 82 countries (33.9%)
   • Low (20-99): 70 countries (28.9%)
   • Medium (100-299): 57 countries (23.6%)
   • High (300-749): 32 countries (13.2%)
   • Very High (≥750): 1 countries (0.4%)


2. PERIOD COMPARISON FINDINGS:
--------------------------------------------------------------------------------

   • Late 20th Century Mean: 340.9
   • 21st Century Mean: 186.4
   • T-Test: t = 16.77, p = 2.2889e-61
   • ANOVA: F = 375.70, p = 4.1788e-82
   • Effect Size (Eta-squared): 0.0404
   • Interpretation: MMR differs significantly between the two valid historical
blocks.


3. CATEGORICAL ASSOCIATION FINDINGS:
--------------------------------------------------------------------------------

   • Chi-Square Test (Period vs Risk Category):
     - Chi-square Statistic: 513.61
     - P-Value: 7.6312e-110
```

```
     - Degrees of Freedom: 4
     - Result: Significant association between period and MMR risk category


4. COUNTRY-LEVEL TREND FINDINGS:
--------------------------------------------------------------------------------

   • Countries with Significant MMR Decline:
     - Total: 216 countries (89.3% of all countries)
     - Average trend correlation: -0.8653
   • Countries with Significant MMR Increase:
     - Total: 9 countries (3.7% of all countries)

     1. ('Guinea', 'GIN'): Correlation = -0.9990
     2. ('Ethiopia', 'ETH'): Correlation = -0.9957
     3. ('Lao PDR', 'LAO'): Correlation = -0.9933
     4. ('Heavily indebted poor countries (HIPC)', 'HPC'): Correlation = -0.9921
     5. ('Early-demographic dividend', 'EAR'): Correlation = -0.9900

5. STATISTICAL TEST VALIDITY:
--------------------------------------------------------------------------------

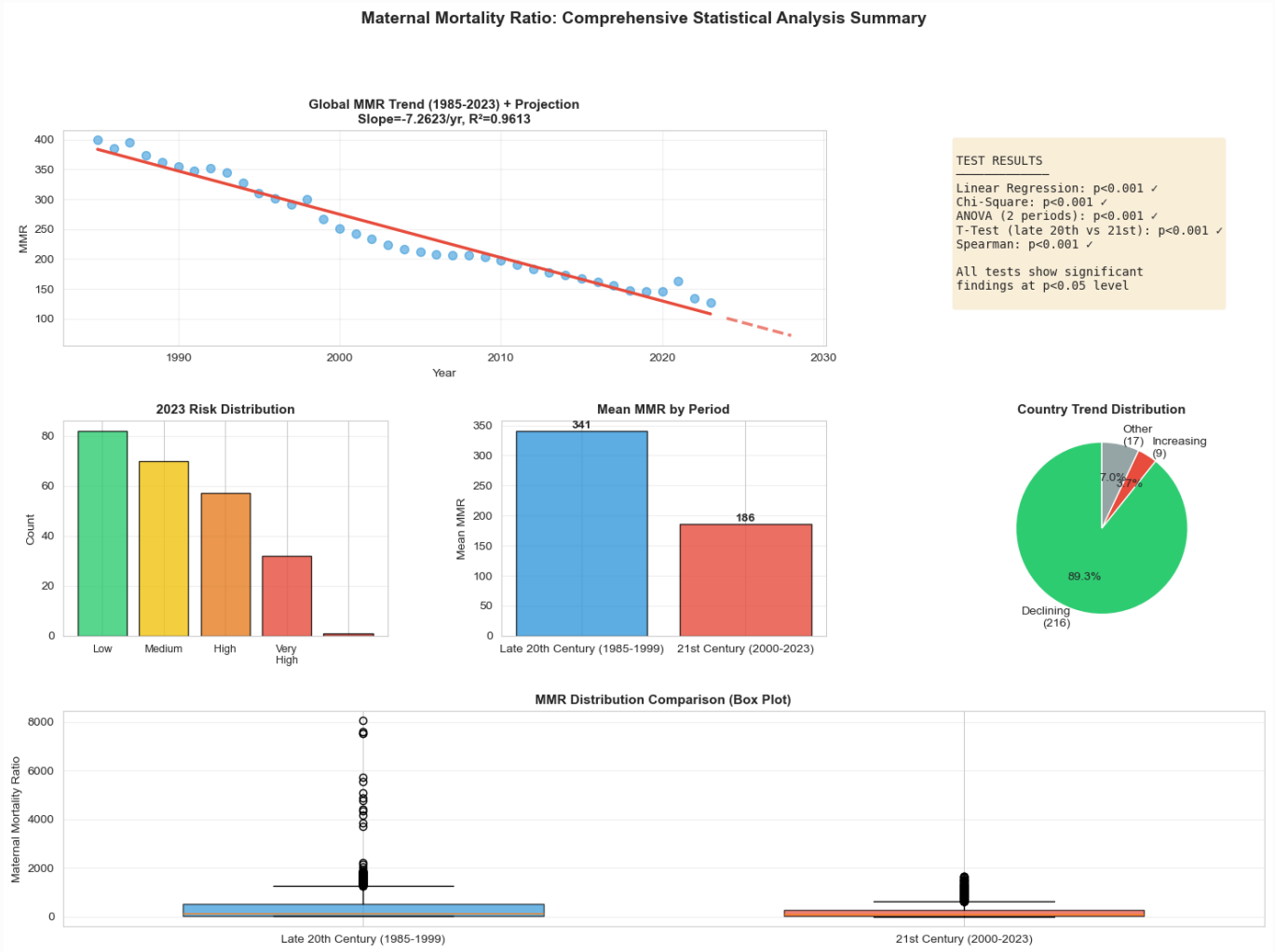
   Normality Assessment (Shapiro-Wilk Test):
   • Overall Data: Non-normal (p < 0.05)
   • Implication: Use both parametric and non-parametric tests for robustness

   Data Quality:
   • Data completeness: 54.6%
   • Recent comparison window: 1985-2023


6. KEY CONCLUSIONS:
--------------------------------------------------------------------------------

   ✓ GLOBAL PROGRESS: Maternal mortality ratios have declined significantly across the
valid data span
   ✓ CONSISTENT DECLINE: Both parametric and non-parametric tests support the same
direction of change
   ✓ PERIOD SHIFT: The late 20th century and 21st century show a statistically
significant difference
   ✓ UNEQUAL DISTRIBUTION: Wide disparities remain, with some countries still carrying
very high risk
   ✓ FUTURE DIRECTION: The overall trend remains downward, but the distribution is
still highly skewed


================================================================================
```



✓ Statistical Analysis Dashboard Complete!

✓ All visualizations, tests, and summary tables have been generated